

Uncrewed Mars Surface Logistics Rover

SYS 625 - Fundamentals of Systems Engineering

Adalberto Lima

SYS 625 Systems Engineer

Connor McGarry

SYS 625 Systems Engineer

Jameson Morris

SYS 625 Systems Engineer

Gowri Das Lovely

SYS 625 Systems Engineer

Revision	Due Date
Mid Term Report	March 26, 2026
Final Report	May 07, 2026

Table of Contents

Section 1. Executive Summary	5
1.1 Mission Statement	5
1.2 Organizations Involved.....	5
1.3 Project Management	5
Section 2. Stakeholder Needs	7
2.1. Stakeholder Overview	7
2.1.1 List of Stakeholders and their Classifications.....	7
2.1.2 Stakeholder Expectations	7
2.2 Potential Stakeholder Conflicts	8
2.2.1 Operational Conflicts	8
2.2.2 Architectural Conflicts.....	9
Section 3. Requirements.....	10
3.1 Stakeholder Elicitation Report	10
3.2 Derived Requirements	14
3.2.1 Stakeholder derived requirements:	14
3.2.2 Fault Derived Sub-System Requirements:.....	17
3.2 Requirements Traceability Matrix.....	19
Section 4. Functional Decomposition	25
4.1 Boundary and Scope.....	25
4.2 Transport Mission Functions (Functional Threads).....	25
4.2.1 Autonomous Navigation under Communication Loss:	25
4.2.2 Mission Tasking and Command Management:	26
4.2.3 Surface Asset Interface and Docking Operations:.....	27
4.2.4 Cargo Handling and Logistics Operations	27
4.2.5 Energy Management and Autonomous Charging:	28
4.2.6 Fault Detection, Isolation and Degraded Mode Operations:.....	29
4.2.7 Communication and Data Integrity Management:	30
4.3 Mode-Based Operational Functions	31
4.4 Interfaces and Dependencies	32
4.4.1 External Interfaces:	32
4.4.2 Internal Dependencies:	32
Section 5. MBSE Artifacts.....	34

5.1 System Context Diagram.....	34
5.2 Functional Architecture	34
5.3 Use Case Diagram/Functional Interactions.....	36
5.4 Requirements Traceability Diagrams.....	37
Section 6. Validation and Verification Plan	39
6.1 Verification Methods	39
6.2 Validation Scenarios under Mars Constraints.....	40
Section 7. System Operations and Sustainment.....	42
7.1 Mission Phases and Modes	42
7.2 Autonomy and Fault Management	46
7.2.1 System P-Diagram.....	46
7.2.2 Failure Modes: Navigation.....	46
7.3 Long-Term Operations and Evolution	48

Figures

Figure 1: System Context Diagram	34
Figure 2: Mars Rover Functional Decomposition	36
Figure 3: Use Case and Functional Interactions	37
Figure 4: Mission Operating Modes	45
Figure 5: System P-Diagram	46
Figure 6: Navigation Failure Response	47
Figure 7: Long-Term User Interactions.....	48

Tables

Table 1: Meeting Attendance	5
Table 2: Section Responsibilities	6
Table 3. Stakeholder Expectations.....	7
Table 4: Derived Requirements.....	14
Table 5: FMEA-Derived Requirements	17
Table 6: Requirements Traceability Matrix	19
Table 7: Verification Methods	39

Acronyms and Abbreviations

MA Mission Assurance

MD Mission Doctrine

NASA National Aeronautics and Space Administration

Ops Operations

P-Diagram..... Parameter Diagram

Section 1. Executive Summary

This section outlines the project's purpose and managerial structure. It highlights the overall intent of the project, identifies key stakeholders, and defines the general roles and responsibilities of each engineer involved. This establishes how decisions are made, and how team members collaborate to achieve the mission objectives.

1.1 Mission Statement

To support sustained human presence on Mars, there is a need for a reliable logistics support systems capable of transporting supplies between landers, habitats, and surface assets. Human-driven logistics are impractical due to risk, time delay, and environmental hazards. This project presents an autonomous logistics rover, emphasizing mission operations, autonomy, safety, and sustainment.

1.2 Organizations Involved

NASA is the primary organization involved in this effort. Within the organization there are subgroups that will be both Active and Passive Stakeholders in the success of the system. One of the stakeholders is the Mission Directorate of the program who will be responsible for the overall success of the system. The Mars surface operations team will be responsible for monitoring the system during its missions. The Habitat and Science Users will be dependent on the system to provide logistical support as outlined in this report. Systems Safety and Mission Assurance Stakeholders oversee that the system will not cause undue harm to any of the active stakeholders. Lastly, the maintainability and sustainment stakeholders are prioritizing the Maintainability and Sustainment goals for the program to ensure the system will serve as a successful logistical solution for many years to come.

1.3 Project Management

Table 1: Meeting Attendance

Date	Meeting Type	Jameson Morris	Gowri Das Lovely	Connor McGarry	Adalbero Lima
2/11/2026	Scheduled	X		X	X
2/25/2026	Scheduled	X	X	X	X
2/26/2026	Ad hoc	X		X	
3/11/2026	Scheduled	X	X	X	X

Date	Meeting Type	Jameson Morris	Gowri Das Lovely	Connor McGarry	Adalbero Lima
3/12/2026	Ad hoc	X	X	X	
3/25/2026	Scheduled	X	X	X	X
4/8/2026	Scheduled	X	X		X
4/22/2026	Scheduled		X	X	X

Table 2: Section Responsibilities

Lead	Section
Jameson Morris	Requirements, Architecture, Traceability
Gowri Das Lovely	Operational Modes, Functional Decomposition, Derived Requirements
Connor McGarry	Requirements, Failure Modes, Architecture
Adalbero Lima	Traceability, Operational Modes, Requirements
Full Systems Team	Document structure and support

Section 2. Stakeholder Needs

This section defines the classification of each key stakeholder and their respective expectations for this project. These expectations are used to identify potential conflicts between stakeholders and resulting design considerations.

2.1. Stakeholder Overview

2.1.1 List of Stakeholders and their Classifications

- NASA (Customer/Active)
- Mars Surface Ops team (Active End User)
- Habitat and Science Users (Active)
- Safety and MA (Passive)
- Maintainability and Sustainment (Active)

2.1.2 Stakeholder Expectations

Table 3. Stakeholder Expectations

Stakeholder	Expectations
NASA	• Reliable cargo transportation without crew involvement • Predictable operations under communications delay • Systems that continue to work after partial failures • Logistics capability that scales with surface operations
Mars Surface Operations Team	• Rover awareness of location and destination • Cargo delivered on time and intact • Confidence that the rover will not block crew operations • Clear operational modes and handoffs
Habitat and Science Users	• Dependable delivery schedule • Reliable payload protection from dust • Reliable payload protection from vibration • Adaptable mating interfaces
Systems Safety and Mission Assurance	• Predictable failure behaviors

	• Hazard mitigation for autonomous navigation • Confidence that the rover will not endanger the crew or habitat
Maintainability and Sustainment	• Increased reliability and minimal maintenance • Modular build to allow refurbishment • Autonomous failure detection • Autonomous self-recovery

2.2 Potential Stakeholder Conflicts

2.2.1 Operational Conflicts

1. Hazard mitigation vs surface asset integration (Habitat/Science users/MA)
 - a. Users want adaptable physical integration, but MA wants a more conservative approach prioritizing safety and redundancy
 - i. Affects
 1. Docking modes
 2. Proximity operations
 3. Speed limits near habitat
 4. Interlocks
2. Working autonomously with partial failures but should arrive on time and intact (MD/Mars Surface Op Team)
 - a. Degraded performance due to partial failures may not be able to protect cargo integrity and deliver to on time
 - b. Fault tolerance may reduce speed or adjust path-finding methods
 - c. System redundancy adds weight/complexity reducing potential cargo payload weight and increasing delivery time
3. Scalability of the system vs overall reliability (MD/MA/Maintenance)
 - a. More rovers or routes
 - b. Overall increase complexity of operations
 - c. Potential for more failure modes from complexity therefore increasing maintenance
4. Cargo protection and delivery efficiency (Surface Ops/Science)
 - a. End user wants payload protected but Surface Ops want on payload delivered on time
 - i. Faster traversal increases potential for payload damage from terrain
 - ii. Slower traversals affect mission duration

2.2.2 Architectural Conflicts

1. Operates with minimal maintenance but needs modularity for future upgrades (MD/Maintainability/MA)
 - a. Upgraded modules can introduce faulty emergent properties
 - i. Software bugs
 - ii. Hardware damage from installation/environment (damage screwing umbilical, introducing dust/sand onto sensitive components)
 1. Overall interface damage
 2. General degradation of system hardware components
2. Autonomy under communication delay vs predictability (MD/Surface Ops/MA)
 - a. Without constant comms, strong autonomy is needed to ensure predictability
 - b. Predictability demands complex and resource-heavy autonomy
 - c. Conflicts between predictable navigation during communication loss and ensuring on-time missions.

Section 3. Requirements

This section lists the needs of the stakeholders along with questions and methods used to create derived requirements. Communication from initial requirements to derivation ensures a means of traceability for Stakeholders and Engineers to reference in design decisions.

3.1 Stakeholder Elicitation Report

1. STAKEHOLDER: MARS SURFACE OPERATIONS TEAM

Number of Needs: 9

UN-MSO-1: We need the rover to know where it is and where it's going."

- Elicitation Method: Use Case Analysis
- Follow-on Question: "In what environment or operational mode must the rover have positional and directional accuracy?"

UN-MSO-1: We need the rover to know where it is and where it's going."

- Elicitation Method: Interview
- Follow-on Question: "What level of position and directional error is acceptable"

UN-MSO-2: We need cargo delivered on time and intact.

- Elicitation Method: Interview
- Follow-on Question: "What time window and cargo integrity would define an on time and intact mission?"

UN-MSO-3: We need confidence the rover will not block crew operations.

- Elicitation Method: Workshops
- Follow-on Question: "Which rover activities might interfere with crew operations?"

UN-MSO-4: We need clear operational modes and handoffs.

- Elicitation Method: Use Case Analysis
- Follow-on Question: "What activities might have unclear operational modes?"

UN-MSO-4: We need clear operational modes and handoffs."

- Elicitation Method: Interview
- Follow-on Question: "What scenarios should rover operations be handed off and who should it be handed off to?"

UN-MSO-5: We need supplies delivered when promised.

- Elicitation Method: Interview
- Follow-on Question: "What type of cargo/supplies should the rover be expected to deliver"

UN-MSO-6: We need hazard mitigation for autonomous navigation."

- Elicitation Method: Use Case Analysis
- Follow-on Question: "What navigational hazards should be detected and mitigated?"

UN-MSO-7: We need assurance the rover cannot endanger crew or habitat.

- Elicitation Method: Workshops
- Follow-on Question: "Under what conditions must the rover restrict or modify its behavior/operations in order to guarantee crew and habitat safety?"

2. STAKEHOLDER: MISSION DIRECTORATE (NASA / SPACE AGENCY)

Number of Needs: 9

UN-MDIR-1: We need reliable cargo transport without crew involvement."

- Elicitation Method: Interview
- Follow-on Question: "What performance metrics would define reliable autonomous cargo transport?"

UN-MDIR-2: We need predictable operations under communication delays.

- Elicitation Method: Workshops
- Follow-on Question: "What type of behavior or operations should the system have under a communication delay?"

UN-MDIR-2: We need predictable operations under communication delays.

- Elicitation Method: Experts
- Follow-on Question: "How long should the system be expected to operate under a communication delay?"

UN-MDIR-3: We need systems that continue working after partial failures.

- Elicitation Method: Interview
- Follow-on Question: "What system or subsystem failures should be considered a partial failure?"

UN-MDIR-3: We need systems that continue working after partial failures.

- Elicitation Method: Experts
- Follow-on Question: "At what point does a partial failure become unacceptable for mission

continuation?"

UN-MDIR-3: We need systems that continue working after partial failures.

- Elicitation Method: Workshops
- Follow-on Question: "What failures would constitute the rover stopping work?"

UN-MDIR-4: We need logistics capability that scales with surface operations.

- Elicitation Method: Documents Review
- Follow-on Question: "What surface operations should the rover support?"

UN-MDIR-5: We need supplies delivered when promised.

- Elicitation Method: Interview
- Follow-on Question: "What type of cargo/supplies should the rover be expected to deliver"

UN-MDIR-6: We need the rover to operate with minimal maintenance.

- Elicitation Method: Experts
- Follow-on Question: "What amount of maintenance is permitted in order to not affect operations?"

3. STAKEHOLDER: HABITAT AND SCIENCE USERS

Number of Needs: 3

UN-MDIR-5: We need supplies delivered when promised.

- Elicitation Method: Interview
- Follow-on Question: "What type of cargo/supplies should the rover be expected to deliver"

UN-HASU-1: We need sensitive payloads protected from dust and vibration.

- Elicitation Method: Experts
- Follow-on Question: "What amount of dust and vibration would be unacceptable to the integrity of sensitive payloads?"

UN-HASU-2: We need the rover to interface with multiple surface assets.

- Elicitation Method: Workshops
- Follow-on Question: "What type of surface assets should the rover be able to interface with?"

4. STAKEHOLDER: SYSTEM SAFETY AND MISSION ASSURANCE

Number of Needs: 3

UN-SSMA-1: We need predictable failure behaviors.

- Elicitation Method: Workshops
- Follow-on Question: "What failure behaviors should be predefined and how should they be classified?"

UN-MSO-6: We need hazard mitigation for autonomous navigation."

- Elicitation Method: Use Case Analysis
- Follow-on Question: "What navigational hazards should be detected and mitigated?"

UN-MSO-7: We need assurance the rover cannot endanger crew or habitat.

- Elicitation Method: Workshops
- Follow-on Question: "Under what conditions must the rover restrict or modify its behavior/operations in order to guarantee crew and habitat safety?"

5. STAKEHOLDER: MAINTAINABILITY AND SUSTAINMENT STAKEHOLDERS

Number of Needs: 4

UN-MDIR-6: We need the rover to operate with minimal maintenance.

- Elicitation Method: Experts
- Follow-on Question: "How often can the rover be maintained in order to not affect missions and operations?"

UN-MASS-1: We need modularity for future upgrades.

- Elicitation Method: Interview
- Follow-on Question: "What type of upgrades would be desired in the future?"

UN-MASS-2: We need fault detection and recovery without human intervention.

- Elicitation Method: Use Case Analysis
- Follow-on Question: "What specific faults must the system recover from completely on its own?"

UN-MASS-2: We need fault detection and recovery without human intervention

- Elicitation Method: Experts
- Follow-on Question: "What defines a successful recovery?"

3.2 Derived Requirements

3.2.1 Stakeholder derived requirements:

Table 4: Derived Requirements

Requirement ID	Requirement Statement	Mission/System
AUTO-001	The fully functioning navigation system shall provide positional accuracy within 10 inches of truth.	System
AUTO-002	The navigation system shall provide directional accuracy within 1 degree of truth.	System
AUTO-003	The navigation system shall avoid commanded keep-out areas.	Mission
AUTO-004	The Rover shall use Image Processing and IMU telemetry for navigation.	System
AUTO-005	The Rover shall maintain a minimum separation distance of 10m from crew while navigating	System
AUTO-006	The Rover shall receive location data from crew	Mission
AUTO-008	The Rover shall automatically stop if a crew member enters the defined safety buffer	Mission
AUTO-009	The Rover shall enter a pre-determined mode if there is a failed self-test for navigation fidelity.	Mission
COM-001	The Rover shall accept commands for areas to avoid.	Mission
COM-002	The Rover shall have a physical switch for crew that will power off the Rover within 5 seconds.	System
COM-003	The Rover shall enter standby mode if no communication has been received within __1hr__.	System
COM-004	The Rover shall receive a command to perform a self-test.	Mission

Requirement ID	Requirement Statement	Mission/System
COM-005	The Rover shall receive a command to go to a target coordinate.	Mission
COM-006	The Rover shall receive a command to end the current task.	Mission
COM-007	The Rover shall send self-test results to operators each time they are performed.	Mission
COM-008	The Rover shall report historical location data since last transmission every 1hr.	System
COM-009	The rover shall self-classify detects faults as Minor, Major, or Critical	Mission
COM-010	The Rover shall support coordination with other rovers	Mission
COM-011	The Rover shall send a request to interface with NASA surface assets before attempting to interface	Mission
COM-012	The Rover shall execute mission tasking following Minor Faults	Mission
COM-013	The Rover shall enter Degraded mode upon Major Faults	Mission
COM-014	The Rover shall enter Safe Mode upon Critical Faults	Mission
OPS-001	The Rover shall be able to charge its batteries by interfacing with charging station.	Mission
PHYS-001	Cargo shall meet condition requirements on delivery defined in Stakeholder documentation.	Mission
PHYS-002	The Rover shall arrive at the target location within 30 minutes of expected time.	Mission
PHYS-003	The Rover shall be able to deliver at least 150 kg of cargo per trip.	Mission
PHYS-004	The Rover shall have a success rate of 95% for arriving at the target location.	Mission

Requirement ID	Requirement Statement	Mission/System
PHYS-005	The Rover shall end the current task and power off in the event of Class A failure.	System
PHYS-006	The Rover shall perform self-tests every 10 hours.	System
PHYS-007	The Rover shall protect cargo by absorbing random vibration intensities of at least 13 Grms.	System
PHYS-008	The Rover shall maintain at least 75 % cargo capacity under degraded power conditions	System
PHYS-009	The Rover shall be able to operate on battery for no less than 5 hours	System
PHYS-010	The Rover shall prevent motion commands while in maintenance mode	Mission
PHYS-011	The Rover shall operate within temperatures of -35°F to +200°F	System
PHYS-012	The Rover shall maintain mobility on slopes up to 60 degrees	System
PHYS-013	The Rover shall tolerate dust exposure levels of IEC 60529 IP6X	System
PHYS-014	The Rover shall have a minimum cargo space of 40 ft^3	System
PHYS-015	The Rover's components shall have a minimum MTBF of 50 years	System
PHYS-016	The Rover's wear components shall have a MTTF of 25 years	System
PHYS-017	The Rover shall have modular cargo area shapes	System
PHYS-018	The Rover Shall Have Modular loading and unloading interfaces	System

3.2.2 Fault Derived Sub-System Requirements:

Subsystem level requirements are derived from the identified failure modes to define how the individual subsystems detect, respond to, and recover from the faults during autonomous operations.

Table 5: FMEA-Derived Requirements

Requirement ID	Subsystem	Requirement Statement	Source Fault
NAV-SUB-001	Navigation (Camera)	The navigation subsystem shall detect camera obstruction by monitoring degradation in image quality or loss of visual features.	Camera Obstruction
NAV-SUB-002	Navigation (Camera)	The navigation subsystem shall command camera reorientation to avoid direct exposure to environmental debris when obstruction is detected.	Camera Obstruction
NAV-SUB-003	Navigation (Camera)	The navigation subsystem flags a navigation fault if visual obstruction persists beyond 5 mins.	Camera Obstruction
SW-SUB-001	Software	The software subsystem shall detect abnormal execution patterns such as Bootlooping using a hardware watchdog timer.	Bootlooping
SW-SUB-002	Software	The software subsystem shall initiate a system reboot when watchdog timeout thresholds are exceeded.	Bootlooping
SW-SUB-003	Software	The software sub system shall limit reboot attempts to 5 consecutive cycles before declaring a persistent fault.	Bootlooping
CDH-SUB-001	Command and Data Handling	The CDH subsystem shall verify integrity of all incoming commands using	Data Corruption

		checksum or Cyclic Redundancy Check (CRC) validation prior to execution.	
CDH-SUB-002	CDH	The CDH subsystem shall reject corrupted commands and prevent execution of invalid data.	Data Corruption
CDH-SUB-003	CDH	The CDH subsystem shall transmit acknowledgement (ACK/NAK) messages based on command validation results.	Data Corruptions
NAV-SUB-004	Navigation (IMU)	The Navigation subsystem shall detect IMU communication loss through the absence of expected sensor data or heartbeat signals.	IMU Connection Loss
NAV-SUB-005	Navigation (IMU)	The navigation subsystem shall attempt to reinitialize IMU communication upon detection of connection loss.	IMU Connection Loss
NAV-SUB-006	Navigation (IMU)	The Navigation subsystem shall flag a hardware fault if IMU communication is not restored after X attempts.	IMU Connection Loss

3.2 Requirements Traceability Matrix

Table 6: Requirements Traceability Matrix

USER NEED ID	Requirement ID	Verification Method
UN-MSO-1: <i>We need the rover to know where it is and where it's going.</i>	AUTO-001	Instrumented Test
	AUTO-002:	Instrumented Test
	COM-005	Demonstration/Field Test
UN-MSO-2: <i>We need cargo delivered on time and intact</i>	AUTO-004	Instrumented Test
	COM-012	Demonstration/Field Test
	PHYS-001	Analysis
	PHYS-002	Demonstration/Field Test
	PHYS-007	Instrumented Test
	PHYS-012	Instrumented Test
	PHYS-013	Instrumented Test
UN-MSO-3 <i>We need confidence the rover will not block crew operations.</i>	AUTO-003	Instrumented Test
	AUTO-005	Instrumented Test
	AUTO-006	Instrumented Test
	AUTO-008	Demonstration/Field Test

USER NEED ID	Requirement ID	Verification Method
UN-MSO-3 <i>We need confidence the rover will not block crew operations.</i>	COM-001	Analysis
UN-MSO-4 <i>We need clear operational modes and handoffs</i>	AUTO-009	Demonstration/Field Test
	COM-002	Demonstration/Field Test
	COM-003	Demonstration/Field Test
	COM-004	Demonstration/Field Test
	COM-011	Demonstration/Field Test
	COM-013	Demonstration/Field Test
	COM-014	Demonstration/Field Test
	OPS-001	Demonstration/Field Test
	PHYS-010	Demonstration/Field Test
UN-MSO-5 <i>We need supplies delivered when promised</i>	AUTO-004	Instrumented Test
	COM-012	Demonstration/Field Test
	PHYS-002	Demonstration/Field Test

USER NEED ID	Requirement ID	Verification Method
UN-MSO-6 <i>We need hazard mitigation for autonomous navigation</i>	AUTO-003	Analysis
	AUTO-004	Demonstration/Field Test
	AUTO-005	Demonstration/Field Test
	AUTO-008	Demonstration/Field Test
UN-MSO-7 <i>We need assurance the rover cannot endanger crew or habitat.</i>	AUTO-005	Demonstration/Field Test
	AUTO-008	Demonstration/Field Test
	COM-001	Simulation
	COM-002	Demonstration/Field Test
	AUTO-003	Demonstration/Field Test
UN-MDIR-1 <i>We need reliable cargo transport without crew involvement.</i>	AUTO-001	Instrumented Test
	AUTO-002	Instrumented Test
	COM-012	Demonstration/Field Test
	PHYS-004	Analysis
	PHYS-006	Demonstration/Field Test
	PHYS-011	Instrumented Test

USER NEED ID	Requirement ID	Verification Method
UN-MDIR-1 <i>We need reliable cargo transport without crew involvement.</i>	PHYS-013	Instrumented Test
UN-MDIR-2 <i>We need predictable operations under communication delays</i>	AUTO-009	Demonstration/Field Test
	COM-003	Demonstration/Field Test
	COM-009	Demonstration/Field Test
	COM-012	Demonstration/Field Test
	COM-013	Demonstration/Field Test
	COM-014	Demonstration/Field Test
	PHYS-006	Demonstration/Field Test
UN-MDIR-3 <i>We need systems that continue working after partial failures.</i>	COM-009	Demonstration/Field Test
	COM-012	Demonstration/Field Test
	COM-013	Demonstration/Field Test
	COM-014	Demonstration/Field Test
UN-MDIR-4 <i>We need logistics capability that scales with surface operations.</i>	COM-001	Analysis
	OPS-001	Demonstration/Field Test
	PHYS-003	Analysis
	PHYS-009	Analysis
	PHYS-014	Inspection

USER NEED ID	Requirement ID	Verification Method
UN-MDIR-5 <i>We need supplies delivered when promised.</i>	PHYS-002	Demonstration/Field Test
	PHYS-004	Analysis
UN-MDIR-6 <i>We need the rover to operate with minimal maintenance.</i>	COM-004	Demonstration/Field Test
	PHYS-006	Demonstration/Field Test
	PHYS-014	Analysis
	PHYS-015	Analysis
UN-HASU-1 <i>We need sensitive payloads protected from dust and vibration.</i>	PHYS-007	Instrumented Test
	PHYS-013	Instrumented Test
UN-HASU-2 <i>We need the rover to interface with multiple surface assets.</i>	COM-011	Demonstration/Field Test
	PHYS-017	Analysis
	PHYS-018	Analysis
UN-SSMA-1 <i>We need predictable failure behaviors</i>	PHYS-014	Analysis
	PHYS-015	Analysis
UN-MASS-1 <i>We need modularity for future upgrades</i>	COM-010	Demonstration/Field Test
	OPS-001	Demonstration/Field Test
	PHYS-017	Analysis
	PHYS-018	Analysis

USER NEED ID	Requirement ID	Verification Method
UN-MASS-2 <i>We need fault detection and recovery without human intervention.</i>	AUTO-009	Demonstration/Field Test
	COM-003	Demonstration/Field Test
	COM-009	Demonstration/Field Test
	COM-012	Demonstration/Field Test
	COM-013	Demonstration/Field Test
	COM-014	Demonstration/Field Test
	PHYS-006	Demonstration/Field Test

Verification methods vary; one example of a requirement that may be verified through testing could be requirement PHYS-011.

“The Rover shall operate within temperatures of -35°F to +200°F”

A test case for this may look like placing the system in a thermal chamber and having the rover perform these basic functionalities within the thermal chamber at the desired temperatures. This test case and requirements could also be flowed down to subsystems and individual components to ensure that there is minimal risk when the full assembled system must be tested.

Not all requirements may be verified through testing, in these cases it is best to verify the requirements via analysis, such as the case of PHYS-015.

“The Rover’s components shall have a minimum MTBF of 50 years”

A mean time between failure is typically verified by analysis due to limitations in simulating the performance of the system over prolonged periods of time. To determine the reliability of the system, a Failure Mode, Effects Critical Analysis (FMECA) is performed on individual parts to estimate how likely a failure is to occur on each individual part. The gathered data is then combined to provide an estimated MTBF for a system that has not been assembled yet.

Section 4. Functional Decomposition

This section provides a detailed breakdown of system functionality within the scope of this report.

4.1 Boundary and Scope

The functional decomposition focuses on the behavior required for an uncrewed rover to autonomously transport cargo between predefined Mars surface assets such as habitats, landers and charging stations.

The scope includes:

- Mission tasking and command handling
- Autonomous navigations
- Docking and interface operations
- Cargo handling and transport
- Communication and data handling
- Energy management
- System health monitoring and fault response

At the highest level, the Rover's primary mission function is to provide autonomous surface logistics of transport between predefined Mars Assets.

4.2 Transport Mission Functions (Functional Threads)

The rover mission is executed through a set of functional threads, where each thread represents a capability that spans multiple subsystems and supports missions' execution.

4.2.1 Autonomous Navigation under Communication Loss:

This describes the rover's ability to navigate autonomously between mission locations when communication with the command center is delayed or temporarily unavailable.

Functions:

1. Determine rover position and orientation
2. Sense environment using onboard sensors (Camera, IMU)
3. Compare sensed environment with onboard maps
4. Plan and update route to destination
5. Avoid hazards and keep out zones
6. Maintain safe distance from other infrastructures.
7. Continue navigation during communication loss

8. Use alternate navigation methods when required.

Subsystems Involved:

1. GNC: Performs localization, path planning, and navigation control.
2. Software and fault management: Supports autonomous decision-making and degraded navigation logic.
3. CDH: Manages navigation data and mission state
4. TTC: Provides communication updates when available.
5. Electrical power: Powers sensing, processing and mobility during traversal
6. Mechanical: Supports structural integration of sensors and mobility hardware.

4.2.2 Mission Tasking and Command Management:

This defines how the rover receives, validates and manages mission commands under delayed or intermittent communication conditions.

Functions:

1. Receives Mission commands and target coordinates
2. Receive keep-out areas and operational constraints
3. Receive self-test and task termination commands
4. Validate incoming commands and data.
5. Store mission st
6. Store mission state during communication delays
7. Synchronize system state with the command center.

Subsystems involved:

1. TTC: Supports uplink/downlink of commands and telemetry
2. CDH: Processes, commands, store mission state and manage command executions.
3. Software and fault management: Validates command logics and manages delayed execution behaviors
4. Harness: Support data transfer between command, communications and onboard control elements.

4.2.3 Surface Asset Interface and Docking Operations:

Here it defines the rover's ability to safely approach and interface with external surface assets.

Functions:

1. Identify target surface assets
2. Enter proximity operations
3. Align rover with docking interface
4. Verify safe approach conditions
5. Establish interface connections
6. Confirm successful docking
7. Detect and handle docking failures.

Subsystems involved:

1. GNC: Supports final alignment and proximity navigation
2. Mobility: Performs controlled approach and positioning
3. Mechanical: Provide structural support for docking and interface stability.
4. Cargo Acquisition, Processing and Handling: Supports interface related positioning for transfer operations.
5. CDH: Coordinates docking sequence logic
6. Software and Fault management: Manages retries, docking failure handling and transition to fault states.
7. TTC: Supports reporting and command updates related to interface operations.

4.2.4 Cargo Handling and Logistics Operations

Here it represents the core logistics capability of the rover.

Functions:

1. Identify and confirm cargo
2. Acquire or secure payload
3. Maintain cargo stability during transport
4. Protect cargo from environmental effects (dust, vibrations and so on)
5. Deliver cargo to destination
6. Release or transfer payload
7. Confirm delivery completion

Subsystems Involved:

1. Payload: Represents the cargo or mission payload being transported.
2. Cargo, Acquisition, Processing and Handling: Performs cargo pickup, storage, release or transfer.
3. Mechanical: Provides the physical structure supporting cargo containment and transfer.
4. Mobility: Transports the payload across the surface
5. Thermal Control: Helps maintain acceptable thermal conditions for sensitive payloads.
6. CDH: Manages cargo state, mission logging, and execution logic.
7. Software & Fault Management: Supports operational sequencing and transfer confirmation.

4.2.5 Energy Management and Autonomous Charging:

Here, the thread ensures sustained rover operation under long-duration mission conditions.

Functions:

1. Monitor battery state and energy usage
2. Determine when charging is required
3. Navigate to charging interface
4. Recharge system
5. Resume operation after charging.
6. Establish charging interface

Subsystems involved:

1. Electrical Power: Monitors state-of-charge, distributes power, and manages charging.
2. CDH: Track system health and power related status information
3. Software & Fault Management: Determines charging transition and mission continuation logic.
4. GNC: Supports navigation to charging assets

5. Mechanical: Executes travel to the charging station
6. TTC: Reports health and charging status when communication is available.

4.2.6 Fault Detection, Isolation and Degraded Mode Operations:

This thread defines how the rover detects, classifies and responds to fault while maintaining safe, and predictable behavior.

Functions:

1. Monitor subsystems health and operational status
2. Detect abnormal behaviors across subsystem
3. Identify and isolate fault source
4. Classify fault severity
5. Execute recovery actions (reset, reboot, retry)
6. Enable assisted motion when required
7. Transition to degraded operation when partial capability is available
8. Transition to safe mode when operation is unsafe
9. Log and report fault conditions.

FAULT CATEGORIES AND HANDLING:

Navigation-related faults:

- Camera obstruction
- IMU Connection loss
- Loss of localization accuracy

Handling approach:

- Detect degradation in navigation data
- Attempt sensor reset or recovery
- Use alternative navigation methods like dead reckoning.
- Transition to Degrade mode if uncertainty increases

Physical and mobility related faults:

- Wheel obstruction or terrain entrapment
- Docking misalignment
- Mechanical interference.

Handling approach:

- Detect abnormal mobility behaviors
- Execute recovery maneuvers
- Enter assisted motion mode
- Escalate to safe mode if recovery fails.

Software and communication related faults:

- Bootlooping or software instability
- Data corruption on receives
- Communication interruption.

Handling approach:

- Detect abnormal execution behavior.
- Validate and reject occupied data
- Restart software process
- Maintain stored mission state during communication loss.

Subsystems involved:

1. Software and fault management: Primary subsystem for health monitoring, fault isolation and recovery actions
2. CDH: Coordinates fault status, logs and mode transition
3. GNC: Supports navigation relates fault responses
4. Mobility: supports assisted motion and recovery from physical obstructions
5. Electrical power: Supports power-related fault detection and safe-state preservation
6. TTC: Transmits fault reports and receives recovery commands when available
7. Mechanical: Maybe implicated in physical failure states and recovery constraints
8. Thermal control: Supports survival and safe operation during degraded or safe mode.

4.2.7 Communication and Data Integrity Management:

This thread defines how the rover maintains reliable communication and data handling.

Functions:

1. Receives commands and system updates
2. Validate incoming data
3. Reject corrupted commands
4. Store mission data for delayed execution
5. Transmit telemetry and system status
6. Handle communication delays or interruption

Subsystems Involved:

1. TTC: Handles communication links and telemetry exchange
2. CDH: Processes, store and routes mission and telemetry data
3. Software and fault management: Supports validation logic and communication fault handling
4. Harness: Supports internal signal transfer between communication and control subsystems
5. Electrical Power: Supports continuous operation of communication and onboard data systems.

4.3 Mode-Based Operational Functions

The rover operates under a set of predefined modes, where each mode activates and prioritizes specific functional threads during mission execution.

1. Idle/Standby Mode: System monitoring, readiness and waiting for mission commands
2. Mission Preparation Mode: Processing mission commands, updating constraints and preparing for execution
3. Self-Navigation Mode: Autonomous traversal, hazard avoidance and position tracking
4. Docking/Interface Mode: Controlled approach, alignment and interface establishment
5. Cargo Transfer Mode: Payload pickup, delivery and transfer confirmation
6. Charging mode: Energy replenishment and systems monitoring
7. Diagnostic/Maintenance Mode: System checks and preparation for fault handling.

Fault-related modes include:

1. Fault scan mode: fault detection and classification
2. Software recovery mode: Rest and recovery of affected systems
3. Assisted motion mode: Recovery from mobility related faults
4. Degraded mode: continued operation with reduced capability

5. Safe mode: Halt operations and maintain critical system functions.

These modes provide structured control over rover behaviors and ensure appropriate system response across nominal and off nominal conditions.

4.4 Interfaces and Dependencies

The rovers functional capabilities depend on coordinated interactions between internal subsystems and external mission elements. The interfaces define how the rover exchange information energy, and physical interactions with its environment, while dependencies describe how functional threads rely on multiple subsystems to operate correctly.

4.4.1 External Interfaces:

The rover interacts with several external subsystems:

- Mission command and control: Provides mission tasking, updates and operational constraints.
- Surface operations and crew systems: Impose safety constraints such as proximity limits and operational boundaries.
- Surface assets (habitats, landers, charging stations): Provide docking, cargo, transfer, and charging interfaces.
- Communication network: Enables command transmission and telemetry reporting under delayed or intermitted conditions
- Martian Environment: Introduces terrain variability, dust and environmental constraints that impact navigation, mobility and systems performance.

These external interfaces directly influence navigation, docking, cargo handling, charging, and fault response behavior.

4.4.2 Internal Dependencies:

Each functional thread depends on coordinated operations across multiple subsystems.

- Navigation functions: Depends on GNC, mobility, sensing, and software control.
- Cargo handling functions: Depends on payload systems, structural support and interface mechanism.
- Docking and interface operations: Depends on navigation accuracy, mobility control, and mechanical alignment.
- Energy management function: Depends on the electrical power subsystem, control logic, and system monitoring.

- Communication functions: Depends on TTC, CDH, and software for data processing and transmission.
- Fault detection and recovery functions: Depends on software and fault management, CDH, and all affected subsystems.

Because of these dependencies, failure or degradation in one subsystem can impact multiple functional threads. This reinforces the need for integrated fault detection and mode-based recovery, as described in earlier sections.

The rover does not operate as isolated subsystems but as an integrated system where functional threads coordinate behavior across hardware and software components. Interfaces enable interaction with external systems, while internal dependencies ensure that mission capabilities such as navigation, cargo transport, and fault response are executed reliably.

Section 5. MBSE Artifacts

This section will provide visual representations of the systems engineering architecture of the Mars Rover.

5.1 System Context Diagram

The system context diagram in Figure 1 illustrates the external interactions with the Mars rover. Received inputs to the Mars rover and expected outputs the system provides to outside entities.

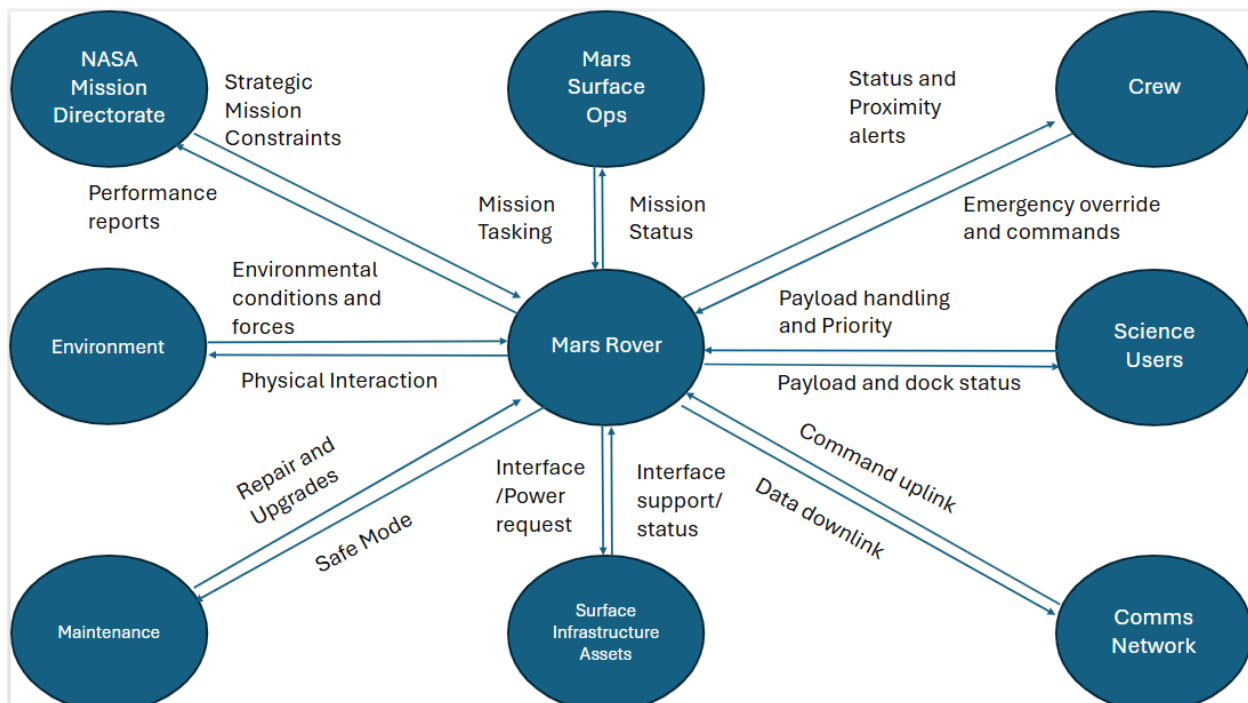


Figure 1: System Context Diagram

5.2 Functional Architecture

The Mars rover is comprised of eleven individual subsystems that work together to perform the mission on the surface of mars. The following subsystems are defined below and illustrated in Figure 2.

Payload: This subsystem is composed of the sensitive payloads that need protected from dust and vibration, such as scientific instruments used in operational concepts such as data collection and cargo analysis

Mechanical: This subsystem is the mechanical structure that mounts internal hardware and contains consuming blocks in a package to protect from the hazardous Martian environment

Telemetry Tracking & Command (TTC): This is the telecommunication subsystem that provides telemetry data such as health data, uplink/downlink sequences used for system command, and tracking instruments and measurements to aid in calculating precise position, velocity, and trajectory.

Thermal Control: This management subsystem actively and passively supports system blocks withstanding the large temperature range (-225F to 70 F) of the mars surface. Active systems would involve a pump circulating liquid throughout the rover to aid in warming or cooling components.

Electrical Power: This subsystem provides continuous power generation to support active and passive modes of operation. This system generates enough power to operate independently for a required amount of time and is capable of being charged after more power-intensive operations.

Mobility: This subsystem is responsible for the propulsion and balance of the rover across terrain of mars.

Harness: This subsystem is a robust network of cables that serve as interconnects between the subsystems of the Mars rover. Capable of distributing power and transmitting data between the subsystems and their components.

Command & Data Handling (CDH): This subsystem is responsible for the command processing, data and communication management, and overall autonomous operation of the Mars rover.

Guidance, Navigation, & Control (GNC): This subsystem uses a combination of sensors and software to determine the vehicle's location, pathfinding, and implementation of getting to a new location or orientation.

Software & Fault Management: This subsystem is comprised of the operating system, artificial intelligence for path finding, autonomous navigation, built-in testing, and mode selection.

Cargo Acquisition, Processing, & Handling: This subsystem is composed of a robotic arm that mechanically acquires scientific samples or other cargo, sensors that analyzes acquired material, and properly stores material for transport

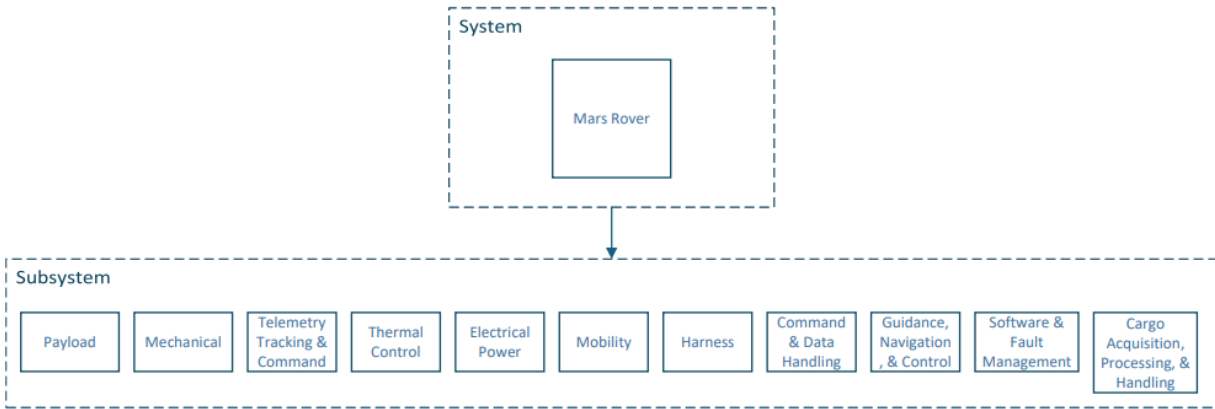


Figure 2: Mars Rover Functional Decomposition

5.3 Use Case Diagram/Functional Interactions

The mars rover is a system to meant to transport supplies or scientific payloads from one location to the next. Figure 3 shows the functional interactions between the subsystems previously defined in Figure 2.

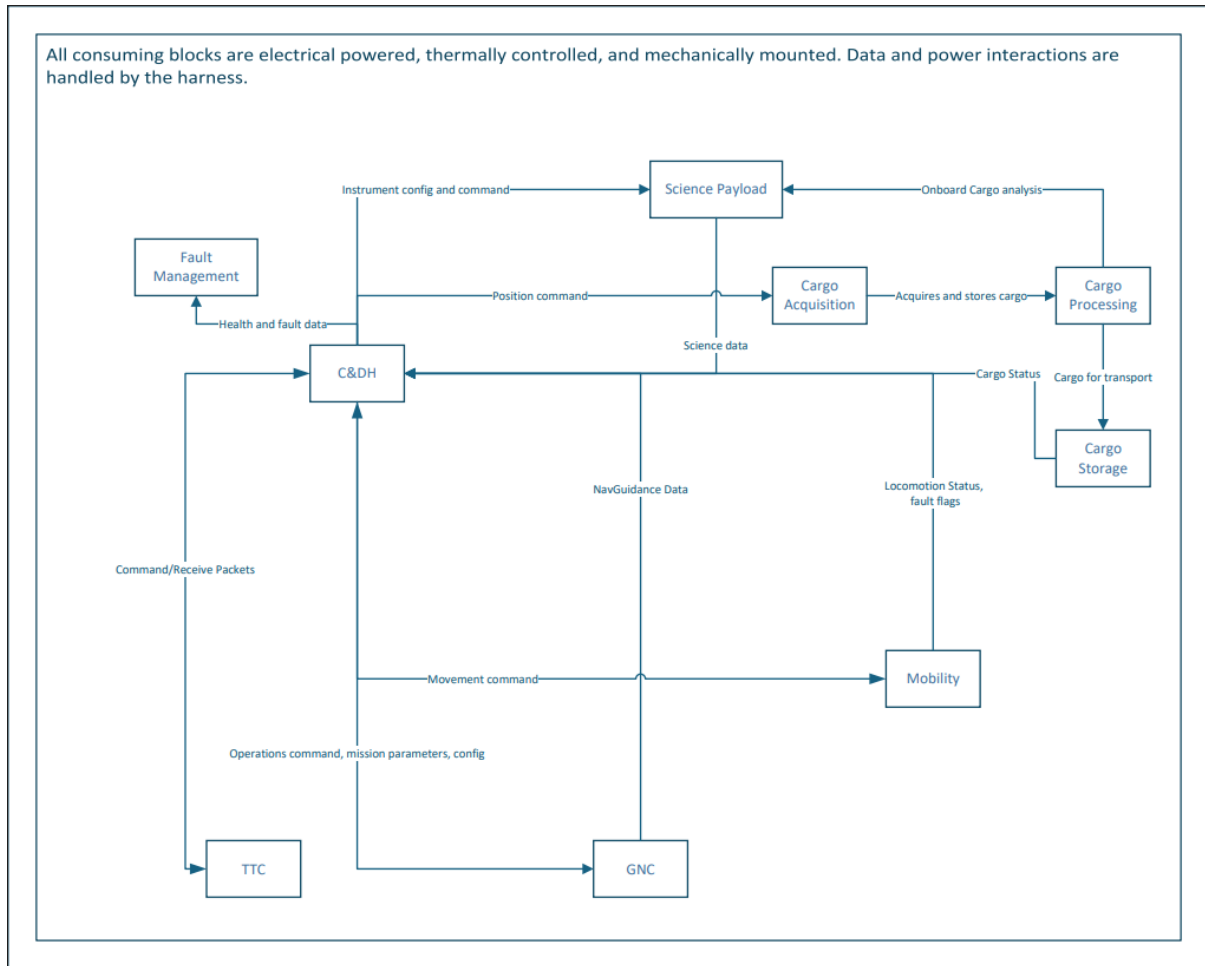


Figure 3: Use Case and Functional Interactions

5.4 Requirements Traceability Diagrams

A requirements traceability diagram was created to illustrate the flow from user need to derived requirements satisfied by subsystem. UN-MSO-1 from the requirements traceability matrix was used as an example in the figure below.

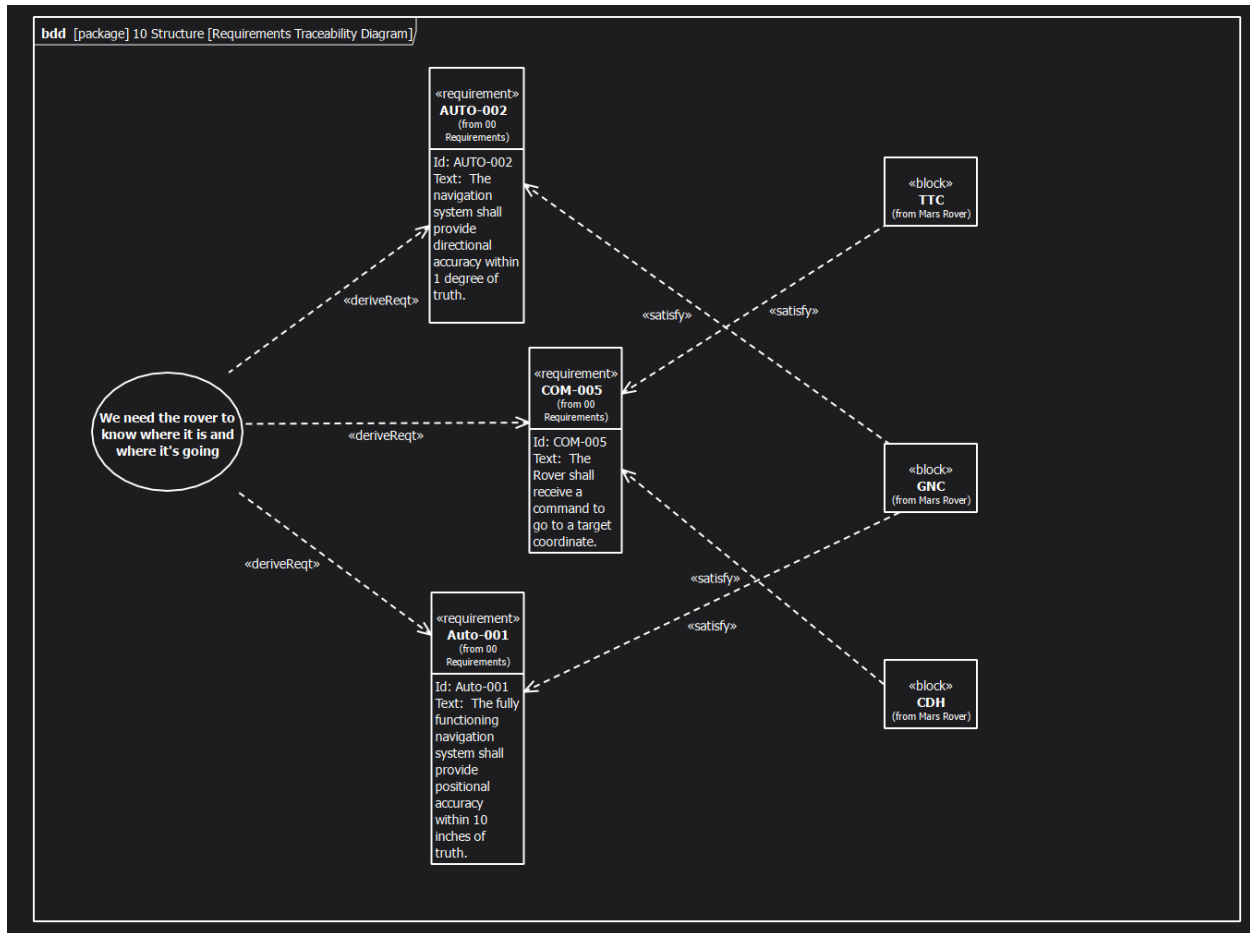


Figure 3: User Need 1 Requirements Traceability Matrix

Section 6. Validation and Verification Plan

This section will outline the plan for verification and validation of the Mars Rover. Included in this section are methods to verify that the Mars Rover meets its requirements as well as validating that the Mars Rover fulfills its stakeholder's needs.

6.1 Verification Methods

The following verification methods are used to confirm that the Mars Rover meets its derived requirements. These methods are aligned with the requirements traceability matrix and are selected based on the type of requirement and the level of testing needed, including the use of fault and Built-In Test (BIT).

Built-In Test (BIT) injections are used to intentionally introduce faults into the system to verify fault detection, classification and correct mode transitions under off-nominal conditions.

Table 7: Verification Methods

Verification Method	Description	Examples/Subsystems
Instrumented Testing	Uses sensors and measurement tools to evaluate system performance under controlled conditions. Fault and BIT injections are applied to intentionally introduce failures and verify fault detection, classification, and system response.	Navigation accuracy testing, environmental tolerance testing, sensor validation, fault injection, BIT injection (e.g., simulate sensor failure, communication loss or obstacle detection to verify fault response and mode transition.)
Demonstration/Field Testing	Involves operating the rover or its subsystems in realistic scenarios to verify functional performance. Subsystems may also be tested independently.	Operate rover, autonomous navigation, docking operations, cargo transfer, isolate and demo subsystems.
Analysis	Uses data, models, and simulations to evaluate system performance and confirm requirement satisfaction.	Finite element analysis of mechanical structure when carrying cargo or navigating uneven terrain. Thermal analysis of whole rover verifies key subsystems like the CDH hardware can survive extreme

		temperature swings on the Martian surface.
Inspection	Reviews physical components, system interfaces, and software to ensure compliance with design requirements.	Physical inspection, interface checks, code reviews.

These methods ensure that both nominal system performance and fault-driven behaviors, including correct mode transitions, are verified across all mission conditions.

6.2 Validation Scenarios under Mars Constraints

Validation focuses on ensuring that the Mars rover fulfills stakeholder needs and performs effectively under realistic Martian environmental and operational conditions. Due to the limitations of testing directly on Mars, validation is primarily conducted using high-fidelity simulation and emulator-based testing environments.

These environments replicate key Mars constraints, including limited communication and latency, dust exposure, rough terrain, thermal extremes, long mission duration and the absence of real-time human control. The rover is evaluated under both nominal and off-nominal scenarios.

Emulator based testing allows engineers to simulate environmental conditions across the upper and lower bounds of what Mars can present, enabling validation of system performance under extreme conditions. In addition, small-scale deployment testing during operations and sustainment (O&S) may be used to validate software updates and system modifications.

The following validation scenarios are used to access system performance.

1. **Autonomous Navigation Under Communication Delay:** The rover operates with delayed or no communication from mission control to validate its ability to navigate using onboard data and maintain mission progress.
2. **Hazard Detection in Dust Condition:** Simulated dust and low visibility conditions are introduced to validate hazard detection and safe navigation.
3. **Fault Detection and Recovery Using BIT Injection:** Artificial faults such as sensor failure or communication loss are introduced to validate fault detection, classification and correct transitions into degrade or safe mode.
4. **Cargo Delivery Under Terrain Constraints:** The rover is tested on uneven terrain to validate timely and safe cargo delivery.

5. Docking and Interface Operations: Docking operations with simulated surface assets are performed to validate alignment accuracy and successful payload transfer.
6. Thermal and Environmental Stress Testing: The rover is exposed to extreme temperature conditions to validate subsystem performance and system survivability.

Section 7. System Operations and Sustainment

This section provides an overview of the modes and operations the Rover performs during its lifetime to fulfill its missions. Failure modes for navigation are analyzed from a software perspective at a high-level to avoid scope outside of SoW. This is meant to give insight as to how the Rover will operate and how it may evolve after continuous operation.

7.1 Mission Phases and Modes

Idle Mode/Standby Mode: The Rover will remain in a low-power standby state while awaiting mission tasks. During this standby state, the rover will undergo periodic system monitoring, battery checks, and environmental status logging. The rover transmits periodic system updates to the command center when communication is available. The rover remains in standby until it receives a mission command from the command center. Once the command and targeted coordinates are received, the rover transitions to the **Self-navigation Mode**.

Self-Navigation Mode: Here the rover navigates autonomously towards the target location. Rover will communicate with the command center every X second(s) and record its latest location in relation to its start and end destination along with its health status. Navigation decisions are made periodically using AI-enhanced image processing algorithms that compare surrounding imagery with onboard maps. IMU telemetry is supplied to the algorithms to aid in analysis and provide logical reasoning for where the Rover is, and where the Rover should be on processing cycle.

If communication with the command center is temporarily unavailable, the rover continues to navigate using stored mission parameters and onboard navigation data. While traveling, the rover will continuously monitor its mobility performance and environmental hazards, and if the rover fails to move or detect an abnormal restriction, the system transitions to **Fault Scan Mode**.

When operating within a defined proximity boundary of the habitat or the crew, the Rover reduces the speed and enforces a safety buffer distance. Once the rover reaches the designated location, it transitions to **Docking/Interface Mode**.

Docking/Interface Mode: When the rover reaches the target location, it transitions into docking mode. During this mode, the rover aligns itself with the designated surface habitat, lander, or a charging station. The rover minimizes its movement during this phase to

ensure stable and safe interface conditions, reducing the risk of misalignment or disturbance to the docking mechanism.

Once properly aligned, the rover establishes a secure mechanical or electrical interface with the surface asset. This interface supports payload handling operations, including the controlled acquisition or release of cargo. Payload transfer is performed through a dedicated mechanism designed to ensure accurate placement, secure attachment and minimal disturbance during exchange.

If the docking attempt fails repeatedly, the rover transitions to **Fault Scan Mode** for diagnostic evaluations and determines the cause of the issue.

Cargo Transfer Mode: Once docking is successfully completed, the Rover initiates the Cargo Transfer Mode. During this mode, the rover confirms cargo integrity before release or pickup and logs in the completion of the transfer. After successful cargo transfer confirmation, the rover transitions back to **Self-navigation Mode** for return travel or transitions **Charging Mode** if battery level requires replenishment.

Charging Mode: If battery level falls below a defined threshold, the Rover transitions into charging mode. During this mode, the mobility commands are disabled while the rover recharges its power system. The rover will continue to monitor its system's health during the charging and transmits status updates when communication is available. Once the sufficient charge is restored, the rover will resume to **Idle Mode** or resume **Self-navigation Mode** depending on the mission tasking.

Diagnostic/ Maintenance mode: This mode entered when the rover detects abnormal system behavior such as mobility failure, sensor inconsistencies, or docking errors. During this mode, it diagnoses the faults, isolates affected subsystems and determines appropriate recovery action before continuing mission operations.

Within diagnostic/maintenance mode, the rover may operate in several sub modes.

Fault Scan Sub-mode: The Rover will perform a diagnostic sweep through all systems and subsystems including mobility, navigation, sensors, power systems, and communications. Subsystem faults are classified into failure classes. Any non-functional subsystems will be trans ponded as a Class A failure. Subsystems with partial functionality will report a Class B failure. All failures are trans ponded to the command center.

Software Recovery Sub-mode: if the fault is determined is related to software or communication anomalies, the rover may attempt automated recovery procedures such as subsystem resets, watchdog-triggered reboots or reinitialization of navigation and communication systems.

Physical Maintenance Sub-mode: if the rover determines that a hardware issue cannot be resolved through software recovery, it may enter a physical maintenance state where motion is restricted, and the system prepared for external intervention. This mode supports future maintenance actions such as module replacement, subsystem upgrades, or remote diagnostic initiated by mission control through communication network.

Depending on the severity of the fault, the rover will either attempt **Assisted motion mode**, or **Degraded mode**, or retry the interruption operation, or transit to **Safe mode** if the fault is deemed critical.

Assisted Motion Mode: This mode is used when the rover is physically obstructed or trapped by terrain features. During this mode, the Rover will momentarily allow the motor to exceed its maximum operating power to dig itself out of a trap. The Rover will also use any additional mechanisms to propel itself if needed, i.e. small thrusters flip over. If the rover successfully frees itself, it will return to **Self-Navigation Mode**. Failure to continue will deploy **Safe Mode**.

Safe Mode: During safe mode the Rover halts all mission activities and enters a protective configuration designed to prevent further system damage. Non-essential subsystems are powered down while critical systems such as communications, thermal control, and fault monitoring remain active. The rover periodically transmits system status signals to mission control to support remote diagnostics. The rover remains in safe mode until recovery commands, system reconfiguration or software updates are transmitted from command center.

Degraded Mode: The rover continues operating with a reduced capability after a non-critical fault. This mode will trigger based on an identified fault during fault-scan mode or with the loss of a specific sensor. While operating in Degraded Mode, the rover may have reduced navigation speed, payload capacity, reporting, or an alternative method of navigation like dead reckoning. The rover continues attempting mission completion while reporting degraded system status to the command center.

The rover transitions between operational modes based on mission commands, environmental conditions, subsystems health status, and fault detection events. Mode transitions are governed by predefined triggers such as failures or completion of cargo transfer operations. These transitions ensure that rover operations remain predictable and safe under the communication delays and environmental constraints of mars surface mission.

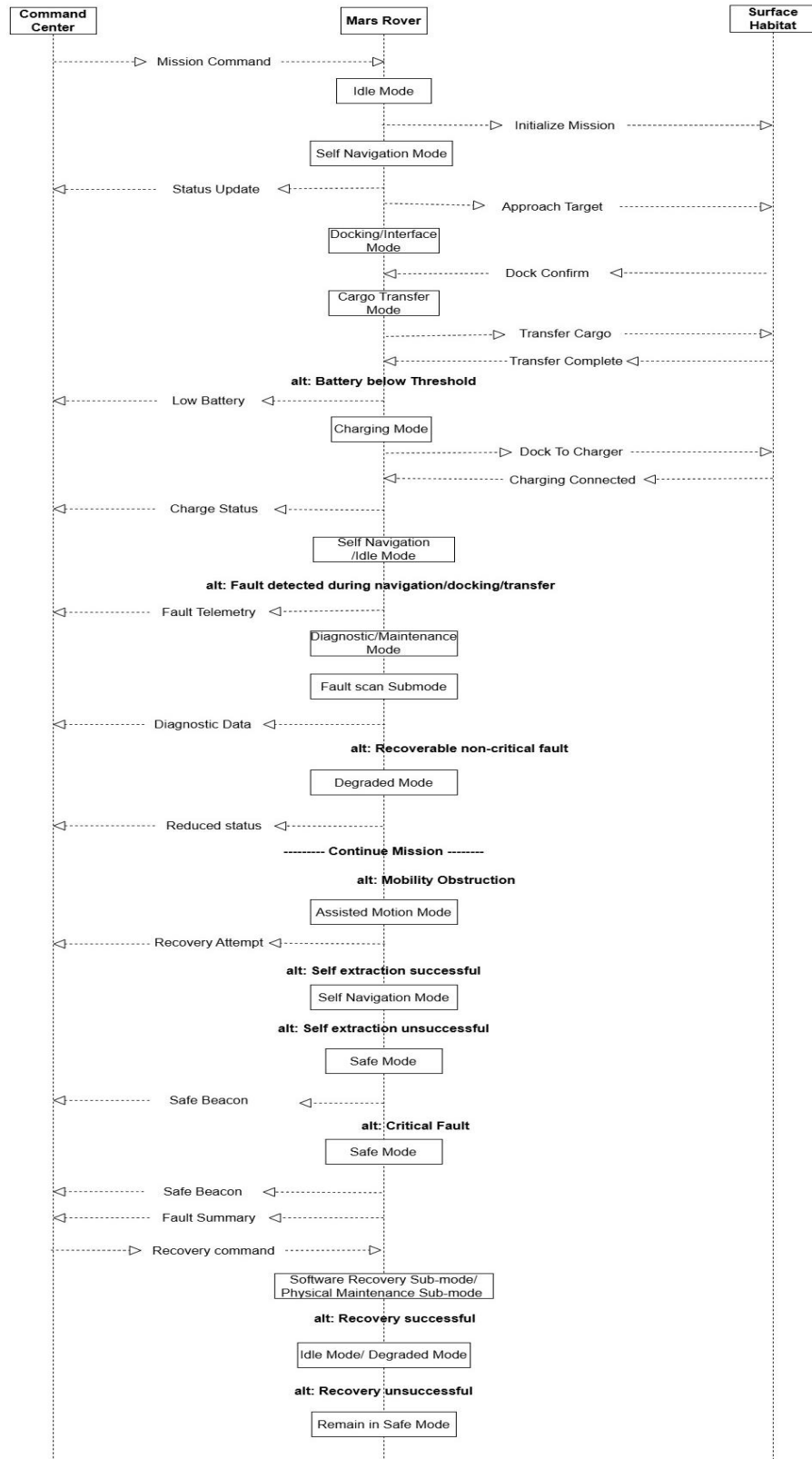


Figure 4: Mission Operating Modes

7.2 Autonomy and Fault Management

7.2.1 System P-Diagram

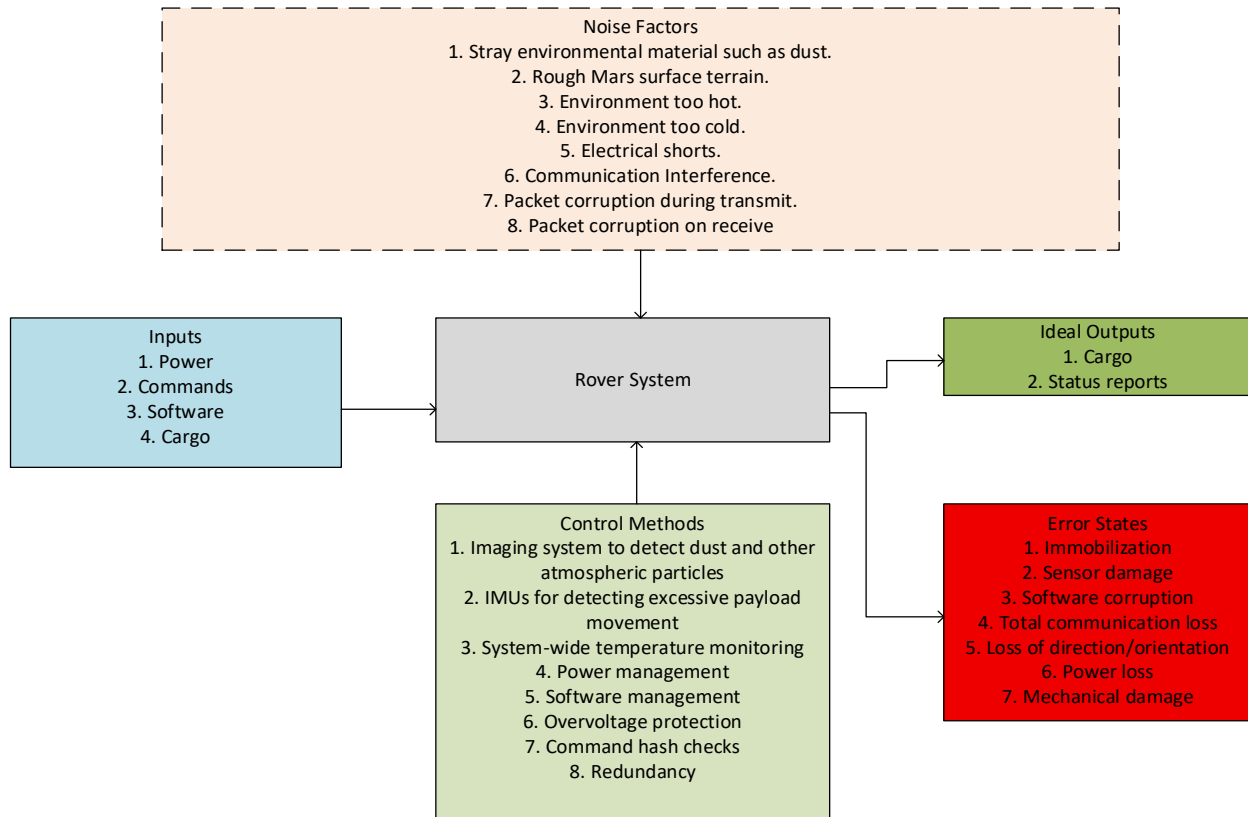


Figure 5: System P-Diagram

7.2.2 Failure Modes: Navigation

Failure Mode	System Response
Camera Obstruction	The Rover will tilt the camera at an angle such that it does not make direct contact with any incoming environmental materials based on sensor readings. This method will be used to passively clear the lens of environmental debris.
Bootlooping	The Rover will employ a Hardware Watchdog timer to detect unexpected software behavior. The Rover will perform a hardware reboot if excessive software errors occur.
Data Corrupted on Receive	The Rover will read a checksum on incoming commands to ensure the data was not corrupted. The rover will

	respond to commands with an ACK or NAK depending on the checksum validation.
Possible IMU Connection Loss	Connection loss with the IMU indicates a potential hardware error. To rule out a serial communication error, the system will transmit a status message and perform a hardware reboot.
Certain IMU Connection Loss	If IMU disconnection persists through a hardware reboot then the system will resort to camera-only navigation. On-board AI will analyze incoming imagery at a greater detail to make up for inaccuracies caused by lack of IMU.

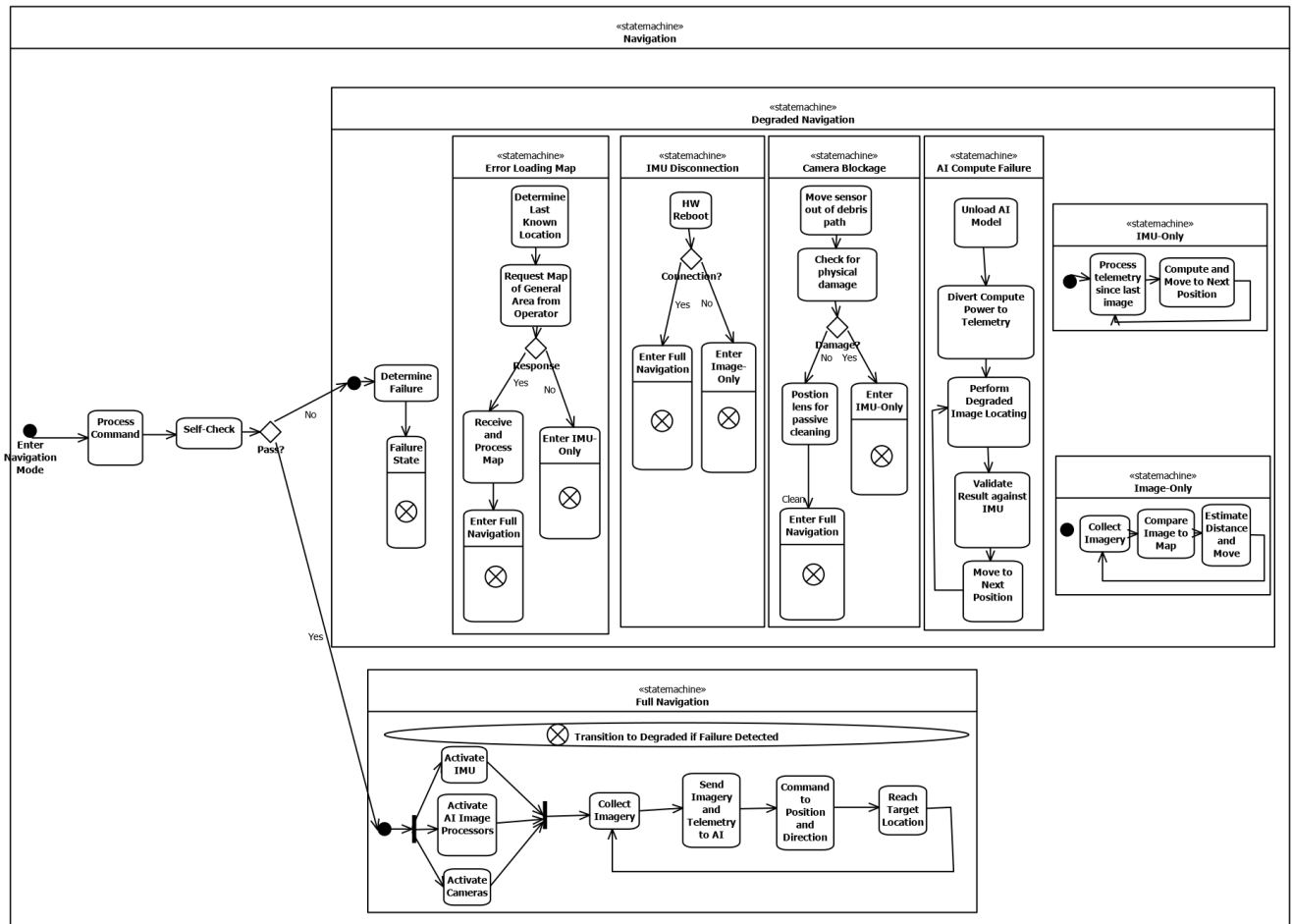


Figure 6: Navigation Failure Response

7.3 Long-Term Operations and Evolution

Long-term operations of the Rover are defined by sustained autonomy, resilience to environmental degradation, and the ability to adapt to changes in CONOPS. Per design, the Rover is intended to maintain reliable performance across extended mission durations while accounting for risk factors by leveraging onboard fault detection, isolation, and recovery to preserve functionality under partial system failures. Operational evolution is enabled through modular hardware interfaces and a configurable software architecture, allowing integration with new surface assets, revised navigation maps, and improved autonomy algorithms as mission needs to change. Per design, the system supports scalable logistics throughput as the number of destinations and demand for cargo delivery increases, while minimizing maintenance through design for reliability and limited crew intervention. Over time, the Rover's capabilities will scale to meet the logistical needs of increasing and continued human presence on Mars.

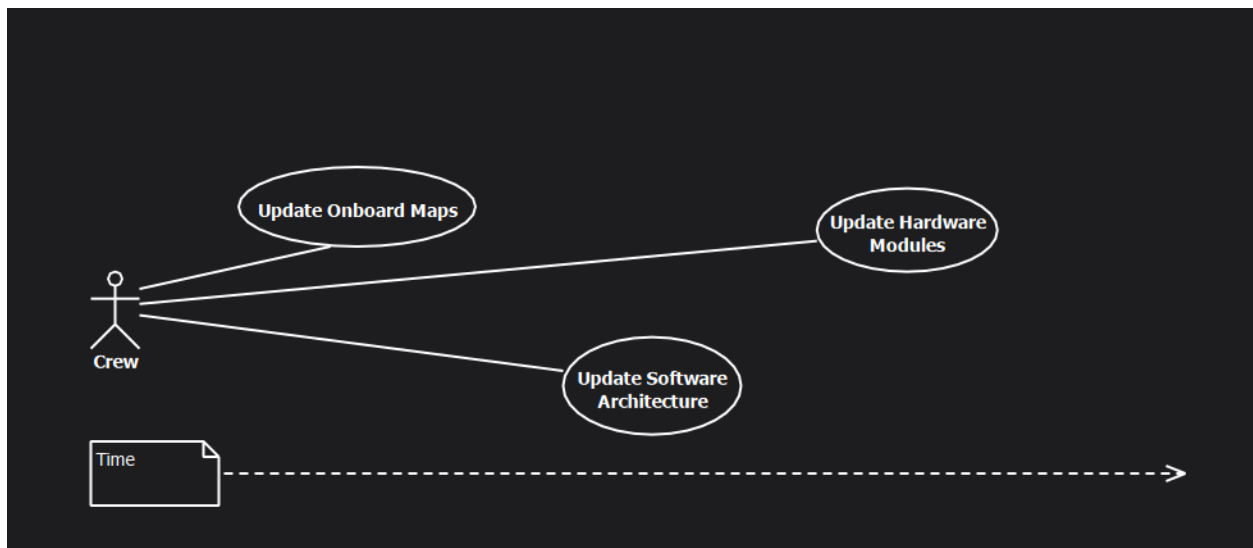


Figure 7: Long-Term User Interactions